

NTRK1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NTRK1 gene fusions from the msk_impact_50k_2026 study, covering 78 fusion events. 32/78 fusions (41.0%) were in-frame. 64/78 fusions (82.1%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.0482628$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for NTRK1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 78 analyzed fusion events, identifying 43 distinct fusion partner genes. The most frequent partner was LMNA, observed in 12 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 30/32 (93.8%) of in-frame fusions retained the domain, $p=0.0482628$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Domain disruption

Domain-disruption analysis was skipped: NTRK1 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 76 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 501 aa (permutation-corrected $p=0.014985$, statistically significant at $\alpha=0.05$). This is -11 aa from the nearest configured domain boundary at 512 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=64$) and "not_retained" ($n=12$). For Frame_status, retained: 30/36 (83.3%, 95% CI 68.1%-92.1%); not_retained: 2/8 (25.0%, 95% CI 7.1%-59.1%). Welch's t-test comparing Tumor_variant_count between groups gave means of 66.3594 vs 43.25, $p=0.200314$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: NTRK1 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
LMNA	12	0.15384615384615385	0.30004340774793187	None	0.851937984496124	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3887286932002887	1
TPM3	12	0.15384615384615385	0.30004340774793187	None	0.7467054263565891	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3689975885491259	2
MEF2D	1	0.01282051282051282	0.30004340774793187	None	1.0	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.36360570572265005	3
TRPM8	1	0.01282051282051282	0.30004340774793187	None	1.0	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.36360570572265005	4
ARHGEF11	2	0.02564102564102564	0.30004340774793187	None	0.9430232558139535	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35773025849545864	5
RABGAP1L	1	0.01282051282051282	0.30004340774793187	None	0.8860465116279069	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3422394266528826	6
VANGL2	1	0.01282051282051282	0.30004340774793187	None	0.8860465116279069	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3422394266528826	7
ZBTB7B	1	0.01282051282051282	0.30004340774793187	None	0.8860465116279069	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3422394266528826	8
SCP2	1	0.01282051282051282	0.30004340774793187	None	0.8651162790697674	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3383150080482314	9
TPR	5	0.0641025641025641	0.30004340774793187	None	0.7427906976744186	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3346097307673727	10
LTAP1	1	0.01282051282051282	0.30004340774793187	None	0.8372093023255813	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3330824499086965	11

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ATP1A2	1	0.01282051282051282	0.30004340774793187	None	0.8186046511627907	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.32959407781567324	12
PLEKHA6	3	0.038461538461538464	0.30004340774793187	None	0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.32874434615198816	13
IRF2BP2	3	0.038461538461538464	0.30004340774793187	None	0.758139534883721	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3278722531287323	14
ANKRD36	1	0.01282051282051282	0.30004340774793187	None	0.8069767441860465	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.32741384525753375	15
RAB25	1	0.01282051282051282	0.30004340774793187	None	0.8046511627906977	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3269777987459059	16
IQGAP3	1	0.01282051282051282	0.30004340774793187	None	0.7976744186046512	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3256696592110221	17
SMYD2	1	0.01282051282051282	0.30004340774793187	None	0.7767441860465116	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.32174524060637094	18
EML4	1	0.01282051282051282	0.30004340774793187	None	0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3191289615366035	19
EPS15	1	0.01282051282051282	0.30004340774793187	None	0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3191289615366035	20
F11R	1	0.01282051282051282	0.30004340774793187	None	0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3191289615366035	21
PEAR1	1	0.01282051282051282	0.30004340774793187	None	0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3191289615366035	22
AFAP1	1	0.01282051282051282	0.30004340774793187	None	0.7488372093023256	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.31651268246683606	23
DIAPH1	1	0.01282051282051282	0.30004340774793187	None	0.7488372093023256	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.31651268246683606	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TRIM63	1	0.01282051282051282	0.30004340774793187	None	0.7488372093023256	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.31651268246683606	25
STK11	1	0.01282051282051282	0.30004340774793187	None	0.7325581395348837	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.31346035688544077	26
KIF21B	1	0.01282051282051282	0.30004340774793187	None	0.7255813953488373	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.312152217350557	27
GON4L	1	0.01282051282051282	0.30004340774793187	None	0.7162790697674419	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3104080313040454	28
BCAN	2	0.02564102564102564	0.30004340774793187	None	0.49534883720930234	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.27379130500708654	29
CTRC	2	0.02564102564102564	0.30004340774793187	None	0.49534883720930234	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.27379130500708654	30
NOS1AP	1	0.01282051282051282	0.30004340774793187	None	0.5186046511627906	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.27334407781567327	31
NELFCD	1	0.01282051282051282	0.30004340774793187	None	0.49534883720930234	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.26898361269939425	32
SEMA4A	1	0.01282051282051282	0.30004340774793187	None	0.49534883720930234	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.26898361269939425	33
COP1	1	0.01282051282051282	0.30004340774793187	None	0.4534883720930233	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2611347754900919	34
TARS2	1	0.01282051282051282	0.30004340774793187	None	0.2697674418604651	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.22668710107148723	35
P2RY8	1	0.01282051282051282	0.30004340774793187	None	0.2581395348837209	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2245068685133477	36
DDR2	1	0.01282051282051282	0.30004340774793187	None	None	0.6202755879498323	['confidence_certainty', 'domain_retention', 'recurrence']	0.21674548396633853	37

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
METTL25B	1	0.01282051282051282	0.30004340774793187	None	None	0.6202755879498323	['confidence_certainty', 'domain_retention', 'recurrence']	0.21674548396633853	38
SLAMF6	3	0.038461538461538464	0.30004340774793187	None	0.11395348837209307	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.20708736940780212	39
CADM3	1	0.01282051282051282	0.30004340774793187	None	0.11395348837209307	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.1974719847924175	40
PRCC	1	0.01282051282051282	0.30004340774793187	None	0.11395348837209307	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.1974719847924175	41
TAF2	1	0.01282051282051282	0.30004340774793187	None	0.11395348837209307	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.1974719847924175	42
SHPRH	1	0.01282051282051282	0.30004340774793187	None	0.0	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.17610570572265005	43

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
71	0	64	1	11	0.0	0.15789473684210525	0.8016323462331666
120	5	59	2	10	0.423728813559322	0.3043106841520683	0.5166827995746912
182	5	59	3	9	0.2542372881355932	0.10748982443269033	0.9686326464637113
187	6	58	3	9	0.3103448275862069	0.14608214986786255	0.83540284831964
284	10	54	5	7	0.25925925925925924	0.05253839246726117	1.2795232195834179
294	11	53	5	7	0.29056603773584905	0.11464025310482523	0.9406628637664377
321	13	51	5	7	0.3568627450980392	0.14136093824402443	0.8496705809825442
343	14	50	5	7	0.392	0.1614921032975089	0.7918487091109843
359	15	49	5	7	0.42857142857142855	0.28166616270038797	0.550265322823117
377	16	48	5	7	0.4666666666666667	0.29450254706111656	0.530910944782467
379	17	47	5	7	0.5063829787234042	0.31247043251825074	0.505191071445305
383	18	46	5	7	0.5478260869565217	0.494013437398657	0.30626123790080845
393	27	37	5	7	1.0216216216216216	1.0	-0.0

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
399	39	25	5	7	2.184	0.33960073354007925	0.46903138034780045
405	39	25	6	6	1.56	0.5328211446391313	0.27341854881920497
411	40	24	6	6	1.6666666666666667	0.5236156499522178	0.28098738162793063
414	41	23	6	6	1.7826086956521738	0.5184932375861125	0.2852569034951671
417	46	18	7	5	1.8253968253968254	0.49401343739865705	0.3062612379008084
418	48	16	7	5	2.142857142857143	0.29450254706111656	0.530910944782467
423	49	15	7	5	2.3333333333333335	0.28166616270038797	0.550265322823117
443	50	14	7	5	2.5510204081632653	0.1614921032975089	0.7918487091109843
452	61	3	8	4	10.166666666666666	0.010191376831302902	1.9917671398557228
492	62	2	8	4	15.5	0.004800736843751964	2.3186920995794114
501	64	0	9	3	None	0.0031294452347083927	2.5045326441976177
571	64	0	10	2	None	0.023157894736842103	1.6353009244666417
616	64	0	11	1	None	0.15789473684210525	0.8016323462331666

domain_retention tables

frame_domain_contingency_table

30	34
2	10

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	76	64	3	9	0.1117283950617284

frequency tables

Partner_gene_counts

Partner_gene	Event_count
AFAP1	1
ANKRD36	1
ARHGEF11	2
ATP1A2	1
BCAN	2
CADM3	1

Partner_gene	Event_count
COP1	1
CTRC	2
DDR2	1
DIAPH1	1
EML4	1
EPS15	1
F11R	1
GON4L	1
IQGAP3	1
IRF2BP2	3
KIF21B	1
LMNA	12
LTAP1	1
MEF2D	1
METTL25B	1
NELFCD	1
NOS1AP	1
P2RY8	1
PEAR1	1
PLEKHA6	3
PRCC	1
RAB25	1
RABGAP1L	1
SCP2	1
SEMA4A	1
SHPRH	1
SLAMF6	3
SMYD2	1
STK11	1
TAF42	1
TARS2	1
TPM3	12
TPR	5
TRIM63	1
TRPM8	1

Partner_gene	Event_count
VANGL2	1
ZBTB7B	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013559-T01-I M5-1	P-0013559-T01-IM5		AFAP1-NTRK1	AFAP1	in-frame	three_prime	156844093	9	393	True	retained	1.0	False	Protein kinase domain			AFAP1(NM_001134647) - NTRK1(NM_002529) Rearrangement : t(1;4)(q23.1;p16.1)(chr1:g.156844093::chr4:g.7852375) Protein fusion: in frame (AFAP1-NTRK1)	NA
EVT-P-0042270-T01-I M6-2	P-0042270-T01-IM6		ANKRD36-NTRK1	ANKRD36	out-of-frame	three_prime	156844593	11	418	True	retained	1.0	False	Protein kinase domain			ANKRD36(NM_001164315) - NTRK1(NM_002529) fusion: t(1;2)(q23.1;q11.2)(chr1:g.156844593::chr2:g.97835462) Protein Fusion: out of frame {ANKRD36:NTRK1}	NA
EVT-P-0017558-T02-I M7-3	P-0017558-T02-IM7		ARHGEF11-NTRK1	ARHGEF11	out-of-frame	three_prime	156845502	12	501	True	retained	1.0	False	Protein kinase domain			ARHGEF11(NM_198236) - NTRK1(NM_002529) fusion: c.32+488:ARHGEF11_c.1501+44:NTRK1inv Protein Fusion: out of frame {ARHGEF11:NTRK1}	NA
EVT-P-0050637-T01-I M6-4	P-0050637-T01-IM6		BCAN-NTRK1	BCAN	in-frame	three_prime	156842373	7	284	True	retained	1.0	False	Protein kinase domain			BCAN(NM_021948) - NTRK1(NM_002529) fusion: c.1064-525:BCAN_c.850+826:NTRK1del Protein Fusion: in frame {BCAN:NTRK1}	NA
EVT-P-0041874-T01-I M6-5	P-0041874-T01-IM6		BCAN-NTRK1	BCAN	in-frame	three_prime	156843258	8	284	True	retained	1.0	False	Protein kinase domain			BCAN(NM_021948) - NTRK1(NM_002529) fusion: c.92-224:BCAN_c.851-167:NTRK1del Protein Fusion: in frame {BCAN:NTRK1}	NA
EVT-P-0055952-T01-I M6-6	P-0055952-T01-IM6		COP1-NTRK1	COP1	in-frame	three_prime	156851106	17	736	True	disrupted	0.17037037037037037	True			Protein kinase domain	RFWD2(NM_022457) - NTRK1(NM_002529) rearrangement: c.2133+18462:RFWD2_c.2206-143:NTRK1inv Protein Fusion: in frame {RFWD2:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 4859-T03-I M6-8	P-0014 859-T0 3-IM6		CTRC- NTRK1	CTRC	unknown	three_prime	156841 593	7	284	True	retained	1.0	False	Protein kinase domain			CTRC (NM_007272) - NTRK1 (NM_002529) rearrangement: c.88:CTRC _c.850+46:NTRK1del Protein Fusion: mid-exon {CTRC:NTRK1}	NA
EVT-P-001 4859-T01-I M6-9	P-0014 859-T0 1-IM6		CTRC- NTRK1	CTRC	unknown	three_prime	156841 592	7	284	True	retained	1.0	False	Protein kinase domain			CTRC (NM_007272) - NTRK1 (NM_002529) rearrangement: c.88:CTRC _c.850+45:NTRK1del Protein fusion: mid-exon (CTRC-NTRK1)	NA
EVT-P-002 9326-T01-I M6-10	P-0029 326-T0 1-IM6		DIAPH 1-NTR K1	DIAPH1	in-frame	three_prime	156843 860	8	393	True	retained	1.0	False	Protein kinase domain			DIAPH1 (NM_005219) - NTRK1 (NM_002529) rearrangement: t(1;5)(q23. 1;q31.3)(chr1:g.156843860 ::chr5:g.140904837) Protein Fusion: in frame {DIAPH1:NTRK1}	NA
EVT-P-005 9139-T01-I M7-11	P-0059 139-T0 1-IM7		EML4- NTRK1	EML4	in-frame	three_prime	156844 256	9	399	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - NTRK1 (NM_002529) fusion: t(1;2)(q23.1;p21)(ch r1:g.156844256::chr2:g.42 473987) Protein Fusion: in frame {EML4:NTRK1}	NA
EVT-P-003 7901-T01-I M6-12	P-0037 901-T0 1-IM6		EPS15 -NTRK 1	EPS15	in-frame	three_prime	156844 212	9	399	True	retained	1.0	False	Protein kinase domain			EPS15 (NM_001981) - NTRK1 (NM_002529) rearrangement: c.2119+12 40:EPS15_c.1195+20:NTR K1inv Protein Fusion: in frame {EPS15:NTRK1}	NA
EVT-P-000 9855-T01-I M5-13	P-0009 855-T0 1-IM5		GON4 L-NTR K1	GON4L	unknown	three_prime	156843 711	8	379	False	retained	1.0	False	Protein kinase domain			GON4L (NM_001037533) - NTRK1 (NM_002529) rearrangement : c.4141:G ON4L_c.1137:NTRK1inv Protein fusion: mid-exon (GON4L-NTRK1)	NA
EVT-P-005 0669-T01-I M6-15	P-0050 669-T0 1-IM6		IQGAP 3-NTR K1	IQGAP3	unknown	three_prime	156844 408	10	414	False	retained	1.0	False	Protein kinase domain			IQGAP3 (NM_178229) - NTRK1 (NM_002529) fusion: c.2531-131:IQGAP 3_c.1241:NTRK1inv Protein Fusion: mid-exon {IQGAP3:NTRK1}	NA
EVT-P-002 2925-T01-I M6-16	P-0022 925-T0 1-IM6		LMNA- NTRK1	LMNA	unknown	three_prime	156844 398	10	411	False	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-3 fused to NTRK1 exons 10 - 17): c.513+1675:LMNA_c. 1231:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 0581-T01-I M5-17	P-0010 581-T0 1-IM5		LMNA- NTRK1	LMNA	in-frame	three_prime	156844 556	10	417	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 11-17): c.51 4-811:LMNA_c.1251+138: NTRK1del Protein fusion: in frame (LMNA-NTRK1)	NA
EVT-P-001 0086-T01-I M5-18	P-0010 086-T0 1-IM5		LMNA- NTRK1	LMNA	in-frame	three_prime	156844 436	10	417	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) rearrangement: c.514-168 6:LMNA_c.1251+18:NTRK 1del Protein fusion: in frame (LMNA-NTRK1)	NA
EVT-P-000 7739-T01-I M5-19	P-0007 739-T0 1-IM5		LMNA- NTRK1	LMNA	unknown	three_prime	156845 035	11	452	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exons 1-8 fused to NTRK1 exons 12-17): c.1443:LMNA_c.13 54+235:NTRK1 Protein fusion: mid-exon (LMNA-NTRK1)	NA
EVT-P-000 3705-T01-I M5-24	P-0003 705-T0 1-IM5		LMNA- NTRK1	LMNA	unknown	three_prime	156845 433	12	492	False	retained	1.0	False	Protein kinase domain			c.686:LMNA_c.1476:NTRK 1del Protein fusion: mid-exon (LMNA-NTRK1)	NA
EVT-P-000 3329-T04-I M6-25	P-0003 329-T0 4-IM6		LMNA- NTRK1	LMNA	unknown	three_prime	156845 200	12	452	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 12 - 17): c.495:LMNA_c.1355-1 12:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1}	NA
EVT-P-000 3329-T03-I M6-26	P-0003 329-T0 3-IM6		LMNA- NTRK1	LMNA	unknown	three_prime	156845 200	12	452	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 12 - 17): c.495:LMNA_c.1355-1 12:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1}	NA
EVT-P-006 6239-T01-I M7-27	P-0066 239-T0 1-IM7		LMNA- NTRK1	LMNA	unknown	three_prime	156845 170	12	452	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1428:LMNA_c.13 55-142:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusion_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0003329-T02-I M6-28	P-0003329-T02-IM6		LMNA-NTRK1	LMNA	unknow_n	thr_ee_pr_im_e	156845200	12	452	True	retain_ed	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exon2 fused with NTRK1 exon12): c.495:LMNA_c.1355-112:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1}	NA
EVT-P-0023994-T06-I M6-29	P-0023994-T06-IM6		LMNA-NTRK1	LMNA	in-fr_ame	thr_ee_pr_im_e	156844525	10	417	True	retain_ed	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1}	NA
EVT-P-0023994-T05-I M6-30	P-0023994-T05-IM6		LMNA-NTRK1	LMNA	in-fr_ame	thr_ee_pr_im_e	156844525	10	417	True	retain_ed	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exon 10 fused with NTRK1 exon 11) : c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1}	NA
EVT-P-0023994-T01-I M6-31	P-0023994-T01-IM6		LMNA-NTRK1	LMNA	in-fr_ame	thr_ee_pr_im_e	156844525	10	417	True	retain_ed	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-10 fused to NTRK1 exons 11-17): c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1}	NA
EVT-P-0025031-T02-I M6-32	P-0025031-T02-IM6		LTAP1-NTRK1	LTA P1	unknow_n	thr_ee_pr_im_e	156846272	14	571	False	disru_p_ted	0.7814814814814814	True			Protein kinase domain	C1orf43 (NM_001098616) - NTRK1 (NM_002529): c.168+114:C1orf43_c.1713:NTRK1inv Protein Fusion: mid-exon {C1orf43:NTRK1}	NA
EVT-P-0024231-T01-I M6-33	P-0024231-T01-IM6		MEF2D-NTRK1	MEF2D	in-fr_ame	five_pr_im_e	156845756	13	501	True	lost	0.0	False		Prot_ein_k_inas_e_do_mai_n		NTRK1 (NM_002529) - MEF2D (NM_005920) rearrangement: c.1502-116:NTRK1_c.1006+1908:MEF2Binv Protein Fusion: in frame {NTRK1:MEF2D}	NA
EVT-P-0045731-T01-I M6-35	P-0045731-T01-IM6		NOS1AP-NTRK1	NOS1A P	unknow_n	thr_ee_pr_im_e	156843455	8	294	False	retain_ed	1.0	False	Protein kinase domain			NOS1AP (NM_014697) - NTRK1 (NM_002529) fusion: c.1106-363:NOS1A_P_c.881:NTRK1dup Protein Fusion: mid-exon {NOS1AP:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 6354-T01-I M6-36	P-0026 354-T0 1-IM6		NTRK1 -ARHGEF11	ARHGEF11	in-frame	three_prime	156845184	12	452	True	retained	1.0	False	Protein kinase domain			ARHGEF11 (NM_198236) - NTRK1 (NM_002529) rearrangement: c.4631-285:ARHGEF11_c.1355-128: NTRK1inv Protein Fusion: in frame {ARHGEF11:NTRK1}	NA
EVT-P-003 5466-T01-I M6-37	P-0035 466-T0 1-IM6		NTRK1 -ATP1A2	ATP1A2	unknown	three_prime	156844714	11	423	False	retained	1.0	False	Protein kinase domain			ATP1A2 (NM_000702) - NTRK1 (NM_002529) rearrangement: c.2563+2: ATP1A2_c.1268:NTRK1dup Protein Fusion: mid-exon {ATP1A2:NTRK1}	NA
EVT-P-005 8090-T04-I M7-38	P-0058 090-T0 4-IM7		NTRK1 -CADM3	CADM3	out-of-frame	five_prime	156836298	4	120	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - CADM3 (NM_021189) rearrangement: c.360-404: NTRK1_c.88+5312:CADM3del Protein Fusion: out of frame {NTRK1:CADM3}	NA
EVT-P-002 1262-T01-I M6-41	P-0021 262-T0 1-IM6		NTRK1 -F11R	F11R	in-frame	three_prime	156844288	10	399	True	retained	1.0	False	Protein kinase domain			F11R (NM_016946) - NTRK1 (NM_002529) rearrangement: c.389-101: F11R_c.1196-75:NTRK1inv Protein Fusion: in frame {F11R:NTRK1}	NA
EVT-P-002 2467-T01-I M6-42	P-0022 467-T0 1-IM6		NTRK1 -IRF2BP2	IRF2BP2	in-frame	three_prime	156844129	9	393	True	retained	1.0	False	Protein kinase domain			IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) fusion (IRF2BP2 exon 1 fused with NTRK1 exons 9-17): c.1001-193:IRF2BP2_c.1178-46:NTRK1inv Protein Fusion: in frame {IRF2BP2:NTRK1}	NA
EVT-P-001 0628-T01-I M5-43	P-0010 628-T0 1-IM5		NTRK1 -IRF2BP2	IRF2BP2	unknown	three_prime	156844224	9	399	True	retained	1.0	False	Protein kinase domain			IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) rearrangement : c.948:IRF2BP2_1195+32:NTRK1inv Protein fusion: mid-exon (IRF2BP2-NTRK1)	NA
EVT-P-001 0628-T03-I M6-44	P-0010 628-T0 3-IM6		NTRK1 -IRF2BP2	IRF2BP2	unknown	three_prime	156844223	9	399	True	retained	1.0	False	Protein kinase domain			IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) rearrangement: c.950:IRF2BP2_c.1195+31:NTRK1inv Protein Fusion: mid-exon {IRF2BP2:NTRK1}	NA
EVT-P-002 7278-T01-I M6-45	P-0027 278-T0 1-IM6		NTRK1 -KIF21B	KIF21B	unknown	three_prime	156843723	8	383	False	retained	1.0	False	Protein kinase domain			KIF21B (NM_001252100) - NTRK1 (NM_002529) rearrangement: c.2077+601:KIF21B_c.1149:NTRK1inv Protein Fusion: mid-exon {KIF21B:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 6197-T01-I M6-46	P-0056 197-T0 1-IM6		NTRK1 -NELF CD	NEL FCD	out-of-frame	five_prime	156843 148	8	284	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - NELFCD (NM_198976) rearrangement: t(1;20)(q23.1;q13.32)(chr1:g.156843148::chr20:g.57565762) Protein Fusion: out of frame {NTRK1:NELFCD}	NA
EVT-P-002 7277-T01-I M6-79	P-0027 277-T0 1-IM6		NTRK1 -PEAR 1	PEAR 1	in-frame	three_prime	156844 219	9	399	True	retained	1.0	False	Protein kinase domain			PEAR1 (NM_001080471) - NTRK1 (NM_002529) fusion (PEAR1 exons 1-15 fused to NTRK1 exons 9-17): c.1951+466:PEAR1_c.1195+27:NTRK1dup Protein Fusion: in frame {PEAR1:NTRK1}	NA
EVT-P-002 9892-T05-I M6-80	P-0029 892-T0 5-IM6		NTRK1 -PLEK HA6	PLE KHA 6	in-frame	three_prime	156844 239	9	399	True	retained	1.0	False	Protein kinase domain			PLEKHA6 (NM_014935) - NTRK1 (NM_002529) fusion: c.2032+105:PLEKHA6_c.1195+47:NTRK1inv Protein Fusion: in frame {PLEKHA6:NTRK1}	NA
EVT-P-002 9892-T04-I M6-81	P-0029 892-T0 4-IM6		NTRK1 -PLEK HA6	PLE KHA 6	in-frame	three_prime	156844 239	9	399	True	retained	1.0	False	Protein kinase domain			PLEKHA6 (NM_014935) - NTRK1 (NM_002529) fusion: c.2032+105:PLEKHA6_c.1195+47:NTRK1inv Protein Fusion: in frame {PLEKHA6:NTRK1}	NA
EVT-P-002 9892-T03-I M6-82	P-0029 892-T0 3-IM6		NTRK1 -PLEK HA6	PLE KHA 6	in-frame	three_prime	156844 239	9	399	True	retained	1.0	False	Protein kinase domain			PLEKHA6 (NM_014935) - NTRK1 (NM_002529) fusion: c.2032+105:PLEKHA6_c.1195+47:NTRK1inv Protein Fusion: in frame {PLEKHA6:NTRK1}	NA
EVT-P-003 8819-T01-I M6-83	P-0038 819-T0 1-IM6		NTRK1 -RAB2 5	RAB 25	out-of-frame	five_prime	156844 458	10	417	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - RAB25 (NM_020387) Rearrangement : Protein Fusion: out of frame {NTRK1:RAB25}	NA
EVT-P-001 5004-T01-I M6-84	P-0015 004-T0 1-IM6		NTRK1 -RABG AP1L	RAB GAP 1L	out-of-frame	five_prime	156844 972	11	452	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - RABGAP1L (NM_014857) rearrangement: c.1354+172:NTRK1_c.2340+50594:RABGAP1Ldel Protein fusion: out of frame (NTRK1-RABGAP1L)	NA
EVT-P-003 8568-T01-I M6-85	P-0038 568-T0 1-IM6		NTRK1 -SEMA 4A	SE MA4 A	out-of-frame	five_prime	156841 963	7	284	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - SEMA4A (NM_001193300) rearrangement: c.850+416:NTRK1_c.301-270:SEMA4Adup Protein Fusion: out of frame {NTRK1:SEMA4A}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 8896-T01-I M6-86	P-0048 896-T0 1-IM6		NTRK1 -SMYD 2	SM YD2	unknown	five_prime	156844 382	10	405	False	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - SMYD2 (NM_020197) rearrangement: c.1215:NTRK1_c.602+876:SMYD2del Protein Fusion: mid-exon {NTRK1:SMYD2}	NA
EVT-P-002 3987-T01-I M6-88	P-0023 987-T0 1-IM6		NTRK1 -TPR	TPR	unknown	three_prime	156844 285	10	399	True	retained	1.0	False	Protein kinase domain			TPR (NM_003292) - NTRK1 (NM_002529) rearrangement: c.2902:TPR_c.1196-78:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA
EVT-P-002 4549-T03-I M6-89	P-0024 549-T0 3-IM6		NTRK1 -TPR	TPR	in-frame	three_prime	156844 290	10	399	True	retained	1.0	False	Protein kinase domain			TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+651_c.1196-73inv Protein Fusion: in frame {TPR:NTRK1}	NA
EVT-P-000 7050-T01-I M5-90	P-0007 050-T0 1-IM5		NTRK1 -VANG L2	VAN GL2	unknown	three_prime	156845 141	12	452	True	retained	1.0	False	Protein kinase domain			VANGL2 (NM_020335) - NTRK1(NM_002529) rearrangement : c.-191+4456:VANGL2_c.1355-171:NTRK1dup Transcript fusion (VANGL2-NTRK1)	NA
EVT-P-000 6172-T01-I M5-91	P-0006 172-T0 1-IM5		P2RY8 -NTRK 1	P2R Y8	unknown	five_prime	156838 012	5	182	False	lost	0.0	False		Protein kinase domain		P2RY8(NM_178129) - NTRK1 (NM_002529) rearrangement: t(X;1)(chrX:g.1598513::chr1:g.156838012) Protein fusion: mid-exon (NTRK1-P2RY8)	NA
EVT-P-004 2526-T02-I M6-92	P-0042 526-T0 2-IM6		PRCC- NTRK1	PRC C	unknown	three_prime	156834 591	3	120	False	retained	1.0	False	Protein kinase domain			PRCC (NM_005973) - NTRK1 (NM_002529) fusion: c.1323+10:PRCC_c.359:NTRK1del Protein Fusion: mid-exon {PRCC:NTRK1}	NA
EVT-P-004 4467-T01-I M6-93	P-0044 467-T0 1-IM6		SCP2- NTRK1	SCP 2	unknown	three_prime	156844 773	11	443	False	retained	1.0	False	Protein kinase domain			SCP2 (NM_002979) - NTRK1 (NM_002529) fusion: c.1338+1979:SCP2_c.1327:NTRK1del Protein Fusion: mid-exon {SCP2:NTRK1}	NA
EVT-P-003 6662-T01-I M6-94	P-0036 662-T0 1-IM6		SHPR H-NTR K1	SHP RH	out-of-frame	five_prime	156831 180	1	71	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - SHPRH (NM_001042683) rearrangement: t(1;6)(q231;q24.3)(chr1:g.156831180::chr6:g.146225530) Protein Fusion: out of frame {NTRK1:SHPRH}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0055621-T05-I M6-95	P-0055621-T05-IM6		SLAMF6-NTRK1	SLAMF6	out-of-frame	threonine_prime	156836411	4	120	True	retained	1.0	False	Protein kinase domain			SLAMF6 (NM_052931) - NTRK1 (NM_002529) fusion: c.50-1837:SLAMF6_c.360-291:NTRK1inv Protein Fusion: out of frame {SLAMF6:NTRK1}	NA
EVT-P-0055621-T02-I M6-96	P-0055621-T02-IM6		SLAMF6-NTRK1	SLAMF6	out-of-frame	threonine_prime	156836411	4	120	True	retained	1.0	False	Protein kinase domain			SLAMF6 (NM_052931) - NTRK1 (NM_002529) rearrangement: c.50-1839:SLAMF6_c.360-291:NTRK1inv Protein Fusion: out of frame {SLAMF6:NTRK1}	NA
EVT-P-0055621-T01-I M6-97	P-0055621-T01-IM6		SLAMF6-NTRK1	SLAMF6	out-of-frame	threonine_prime	156836411	4	120	True	retained	1.0	False	Protein kinase domain			SLAMF6 (NM_052931) - NTRK1 (NM_002529) rearrangement: c.50-1839:SLAMF6_c.360-291:NTRK1inv Protein Fusion: out of frame {SLAMF6:NTRK1}	NA
EVT-P-0022936-T01-I M6-98	P-0022936-T01-IM6		STK11-NTRK1	STK11	unknown	five_prime	156848955	15	616	False	disrupted	0.3888888888888889	True			Protein kinase domain	NTRK1 (NM_002529) - STK11 (NM_000455) rearrangement: t(1;19)(;)(c chr1:g.56848955::chr19:g.204186) Protein Fusion: mid-exon {NTRK1:STK11}	NA
EVT-P-0024875-T01-I M6-99	P-0024875-T01-IM6		TAF12-NTRK1	TAF12	unknown	threonine_prime	156836548	4	120	True	retained	1.0	False	Protein kinase domain			NTRK1 (NM_002529) rearrangement: t(1;12)(q23.1;q14.1)(chr1:g.156836548::chr12:g.62314748) Transcript Fusion {FAM19A2:NTRK1}	NA
EVT-P-0069757-T01-I M7-100	P-0069757-T01-IM7		TARS2-NTRK1	TARS2	unknown	threonine_prime	156838028	5	187	False	retained	1.0	False	Protein kinase domain			TARS2 (NM_025150) - NTRK1 (NM_002529) fusion: c.264-188:TARS2_c.561:NTRK1del Protein Fusion: mid-exon {TARS2:NTRK1}	NA
EVT-P-0050767-T01-I M6-101	P-0050767-T01-IM6		TPM3-NTRK1	TPM3	in-frame	threonine_prime	156845252	12	452	True	retained	1.0	False	Protein kinase domain			TPM3(NM_153469) - NTRK1 (NM_002529) Fusion(TPM3 exon 12 with NTRK1 exon12): c.665-654:TPM3_c.1355-60:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-0049078-T04-I M6-102	P-0049078-T04-IM6		TPM3-NTRK1	TPM3	unknown	threonine_prime	156843537	8	321	False	retained	1.0	False	Protein kinase domain			TPM3 (NM_001278190) - NTRK1 (NM_002529) fusion: c.395-6179:TPM3_c.963:NTRK1inv Protein Fusion: mid-exon {TPM3:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 9078-T01-I M6-103	P-0049 078-T0 1-IM6		TPM3- NTRK1	TPM 3	unknown	thre e_ pr im e	156843 537	8	321	False	retai ned	1.0	False	Protein kinase domai n			TPM3 (NM_001278190) - NTRK1 (NM_002529) fusion: c.395-6179:TPM3_ c.963:NTRK1inv Protein Fusion: mid-exon {TPM3:NTRK1}	NA
EVT-P-004 6877-T02-I M6-104	P-0046 877-T0 2-IM6		TPM3- NTRK1	TPM 3	in-fr ame	thre e_ pr im e	156843 811	8	393	True	retai ned	1.0	False	Protein kinase domai n			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.776-357:TPM3_ c.1177+60:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-002 6375-T04-I M6-105	P-0026 375-T0 4-IM6		TPM3- NTRK1	TPM 3	in-fr ame	thre e_ pr im e	156844 046	9	393	True	retai ned	1.0	False	Protein kinase domai n			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.775+355:TPM3_ c.1178-129:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-002 6375-T01-I M6-106	P-0026 375-T0 1-IM6		TPM3- NTRK1	TPM 3	in-fr ame	thre e_ pr im e	156844 046	9	393	True	retai ned	1.0	False	Protein kinase domai n			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 9-17): c.775+355:TPM3_ c.1178-129:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-002 5815-T02-I M6-107	P-0025 815-T0 2-IM6		TPM3- NTRK1	TPM 3	in-fr ame	thre e_ pr im e	156844 230	9	399	True	retai ned	1.0	False	Protein kinase domai n			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 with NTRK1 exons 10-17): c.*6032:TPM3_c.1195+38: NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-002 1467-T01-I M6-108	P-0021 467-T0 1-IM6		TPM3- NTRK1	TPM 3	in-fr ame	thre e_ pr im e	156844 038	9	393	True	retai ned	1.0	False	Protein kinase domai n			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 fused in-frame to NTRK1 exons 9-17): c.*652:TPM3_ c.1178-137:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-002 1170-T03-I M6-109	P-0021 170-T0 3-IM6		TPM3- NTRK1	TPM 3	in-fr ame	thre e_ pr im e	156845 218	12	452	True	retai ned	1.0	False	Protein kinase domai n			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_ c.1355-94:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA

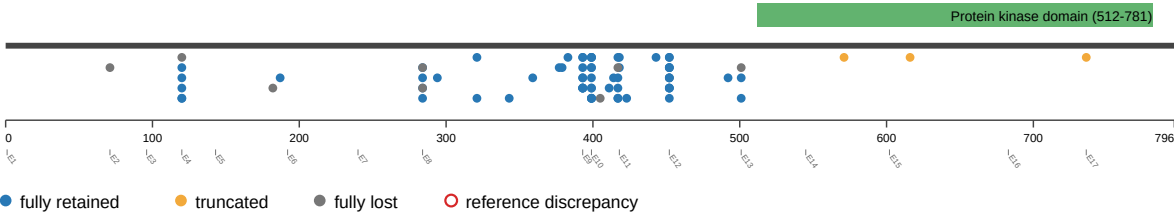
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 1170-T02-I M6-110	P-0021 170-T0 2-IM6		TPM3- NTRK1	TPM3	in-frame	three_prime	156845218	12	452	True	retained	1.0	False	Protein kinase domain			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_c.1355-94:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-001 4143-T01-I M5-111	P-0014 143-T0 1-IM5		TPM3- NTRK1	TPM3	in-frame	three_prime	156844002	9	393	True	retained	1.0	False	Protein kinase domain			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-10 fused in-frame to NTRK1 exons 9-17): c.*1568:TPM3_c.1178-173:NTRK1inv Protein fusion: in frame (TPM3-NTRK1)	NA
EVT-P-006 9266-T01-I M7-112	P-0069 266-T0 1-IM7		TPM3- NTRK1	TPM3	unknown	three_prime	156843601	8	343	False	retained	1.0	False	Protein kinase domain			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.854+282:TPM3_c.1027:NTRK1inv Protein Fusion: mid-exon {TPM3:NTRK1}	NA
EVT-P-004 9951-T01-I M6-113	P-0049 951-T0 1-IM6		TPR-N TRK1	TPR	in-frame	three_prime	156844660	11	418	True	retained	1.0	False	Protein kinase domain			TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2610+187:TPR_c.1252-38:NTRK1inv Protein Fusion: in frame {TPR:NTRK1}	NA
EVT-P-004 2965-T01-I M6-114	P-0042 965-T0 1-IM6		TPR-N TRK1	TPR	unknown	three_prime	156843703	8	377	False	retained	1.0	False	Protein kinase domain			TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531:TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA
EVT-P-006 5034-T02-I M7-115	P-0065 034-T0 2-IM7		TPR-N TRK1	TPR	unknown	three_prime	156843651	8	359	False	retained	1.0	False	Protein kinase domain			TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2777-512:TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA
EVT-P-001 6347-T01-I M6-116	P-0016 347-T0 1-IM6		TRIM6 3-NTRK1	TRIM63	in-frame	three_prime	156844048	9	393	True	retained	1.0	False	Protein kinase domain			TRIM63 (NM_032588) - NTRK1 (NM_002529) Rearrangement : c.1051+386:TRIM63_c.1178-127:NTRK1inv Protein Fusion: in frame {TRIM63:NTRK1}	NA
EVT-P-000 7755-T02-I M5-117	P-0007 755-T0 2-IM5		TRPM8-NTRK1	TRPM8	out-of-frame	three_prime	156845664	12	501	True	retained	1.0	False	Protein kinase domain			TRPM8 (NM_024080) - NTRK1 (NM_002529) rearrangement: t(1;2)(q23.1; q27.1)(chr1:g.156845664::chr2:g.234866124) Protein fusion: out of frame (TRPM8-NTRK1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000104-T02-IM6-118	P-0000104-T02-IM6		ZBTB7B-NTRK1	ZBTB7B	unknown	three_prime	156845173	12	452	True	retained	1.0	False	Protein kinase domain			ZBTB7B (NM_001252406) - NTRK1 (NM_002529) Rearrangement : c.944:ZBTB7B_c.1355-139:NTRK1 del Protein fusion: mid-exon (ZBTB7B-NTRK1)	NA

Figures

domain retention outliers

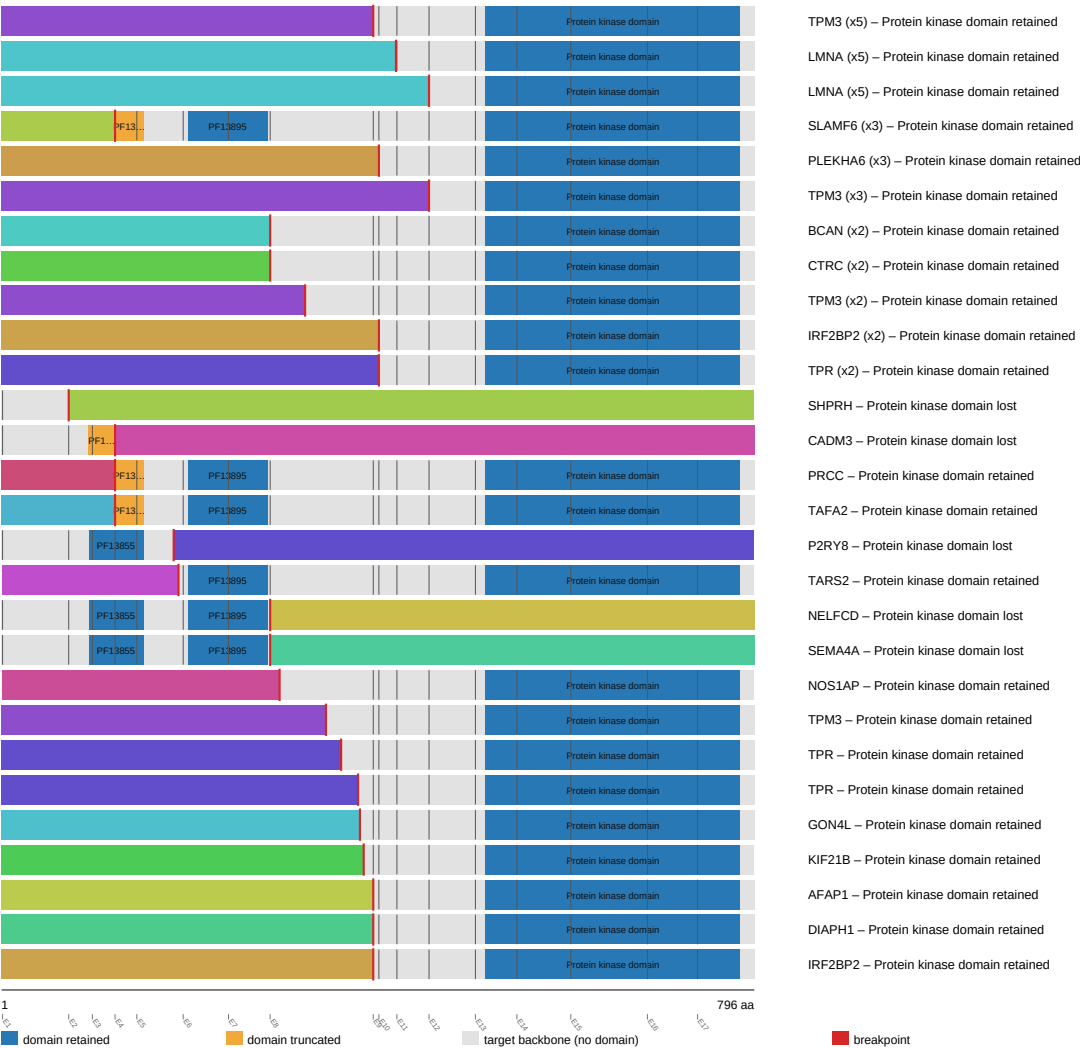
NTRK1 domain-retention track



fusion schematic

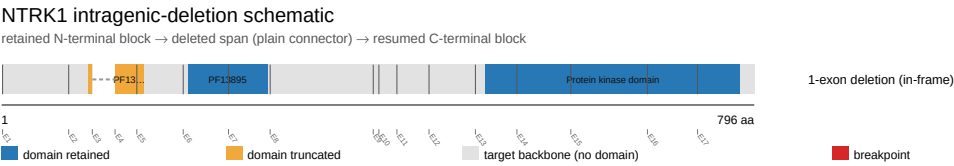
NTRK1 fusion-transcript schematic

partner block → breakpoint → retained NTRK1 portion (domain-colored, exon ticks)



Showing the top 28 of 53 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

